

LOCARNO-MONTI, Switzerland

WMO: 067600

Lat: 46.1722N

Lon: 8.7875E

Elev: 380

StdP: 96.84

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	-1.2	-16.8	0.9	3.6	-14.2	1.1	4.9	5.4	7.2	4.7	7.5	1.1	10	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	30.5	20.7	29.1	20.1	27.7	19.4	21.9	28.8	21.1	27.5	20.4	26.2	2.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	15.0	25.0	18.9	14.3	24.4	18.2	13.7	23.7	65.8	28.9	62.8	27.6	60.2	26.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.2	3.4	-4.2	32.5	2.1	1.8	-5.8	33.8	-7.0	34.9	-8.2	35.9	-9.7	37.2		
(5)			-6.0	23.2	1.8	0.8	-7.3	23.8	-8.3	24.2	-9.3	24.6	-10.6	25.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	4.2	5.5	9.7	12.9	16.6	20.7	22.6	22.0	17.9	13.5	8.3	4.8		
	DBStd	7.16	2.75	3.18	3.62	3.51	3.22	3.19	2.64	2.62	2.73	2.89	2.91	2.86		
	HDD10.0	611	183	129	51	12	1	0	0	0	6	66	163			
	HDD18.3	2236	439	359	269	165	71	14	3	38	151	302	421			
	CDD10.0	1801	2	3	39	99	207	319	392	371	238	114	14	1		
	CDD18.3	386	0	0	1	2	18	84	137	116	27	1	0	0		
	CDH23.3	2316	0	0	3	12	89	523	915	678	91	4	0	0		
	CDH26.7	545	0	0	0	2	10	124	235	167	7	1	0	0		
(14) Wind	WSAvg	1.3	1.0	1.2	1.4	1.5	1.5	1.5	1.4	1.3	1.2	1.1	1.0	0.9		
(15) Precipitation	PrecAvg	1726	73	67	81	148	194	180	167	178	190	180	194	74		
	PrecMax	2559	244	278	277	400	553	447	325	319	442	576	742	225		
	PrecMin	993	0	4	6	13	71	27	31	21	20	10	3	0		
	PrecStd	378	62	62	63	97	104	94	80	74	130	122	176	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	17.2	22.4	24.8	28.0	31.7	32.2	32.8	27.6	23.3	17.1	14.3		
		MCWB	7.9	8.0	10.9	14.2	17.7	20.6	21.6	21.9	19.4	15.1	10.0	6.2		
	2%	DB	11.7	13.5	19.1	22.3	25.7	29.6	30.5	30.2	25.7	20.6	14.8	11.5		
		MCWB	5.4	6.5	9.8	13.0	16.4	20.3	20.8	20.7	18.2	14.8	9.3	5.6		
	5%	DB	9.5	11.7	17.1	20.7	23.9	28.0	29.1	28.3	24.1	19.1	13.3	9.7		
		MCWB	4.3	5.6	9.1	11.9	15.5	19.6	20.3	19.9	17.3	13.8	9.0	5.3		
	10%	DB	7.9	10.1	15.3	18.8	22.2	26.3	27.8	26.9	22.6	17.6	12.0	8.4		
		MCWB	3.8	4.8	8.2	11.2	14.7	18.7	19.5	19.2	16.4	13.5	8.8	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	8.9	12.0	15.4	19.0	22.6	23.2	22.7	20.6	17.1	12.6	9.2		
		MCDB	14.9	15.1	19.0	22.7	26.4	29.5	30.0	30.4	26.2	20.3	13.9	10.7		
	2%	WB	6.6	7.4	10.9	14.0	17.6	21.3	22.1	21.7	19.3	16.1	11.4	7.6		
		MCDB	9.8	11.8	16.9	20.9	24.0	27.9	28.9	28.6	24.1	18.7	12.9	9.1		
	5%	WB	5.6	6.4	10.0	13.0	16.6	20.3	21.2	20.9	18.5	15.3	10.6	6.6		
		MCDB	8.2	10.3	15.2	19.0	22.2	26.6	27.6	26.9	22.5	17.7	12.0	8.5		
	10%	WB	4.6	5.6	9.1	12.1	15.7	19.4	20.4	20.2	17.6	14.5	9.7	5.7		
		MCDB	7.0	9.0	13.7	17.0	20.6	25.2	26.4	25.5	21.3	16.8	11.2	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.4	7.9	8.0	8.1	8.5	8.9	8.5	7.4	6.0	5.1	4.8		
		MCDBR	8.0	8.8	10.5	10.8	10.7	10.4	10.4	10.3	9.3	8.1	7.0	6.6		
	5% WB	MCWBR	4.7	5.1	5.5	5.4	5.2	4.9	4.7	4.6	4.6	4.4	4.2	4.2		
		MCWBR	6.5	7.7	9.0	9.4	9.2	9.4	9.5	9.1	8.0	6.2	4.6	5.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.307	0.341	0.400	0.465	0.478	0.495	0.480	0.470	0.447	0.411	0.348	0.303		
		taud	2.299	2.218	2.108	1.992	2.037	2.053	2.108	2.144	2.166	2.216	2.325	2.347		
	Ebn at Noon	Ebn at Noon	779	816	813	791	796	783	790	780	759	724	719	742		
		Edh at Noon	78	104	135	166	164	163	152	142	127	105	77	68		
(44) All-Sky Solar Radiation	RadAvg	1.56	2.47	3.73	4.48	5.03	5.88	5.92	5.08	3.91	2.49	1.52	1.30			
	RadStd	0.20	0.31	0.52	0.66	0.47	0.46	0.46	0.40	0.34	0.33	0.24	0.17			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.43	N/A	N/A	+0.84	N/A	N/A	N/A	N/A	+70	+70
(47) Regional (116 neighbors)		N/A	N/A	N/A	+0.72	+0.40	+0.30	N/A	N/A	+58	+16

Nomenclature: See separate page